Google Cloud H2S

Gen AI Academy

APAC Edition

Build in APAC. Build for the world.

Participant Details

a. Participant name: Omanand Swami
b. Problem Statement:Travel information is scattered across multiple sources, tourist guides are static and outdated, LLMs hallucinate travel facts, and there is no standardised way for AI agents to call external tools and APIs in real-time.

Brief about the idea

Travel Guide Agent is an AI-powered travel assistant built using G**oogle ADK and Model Context Protocol (MCP).** It uses **Gemini 2.5 Flash** to answer natural language questions about any country returning real-time facts (capital, population, currency, language), rich Wikipedia summaries, and region-aware travel advisories. The agent connects to live **REST APIs via a custom MCP server**, eliminating hallucinations and providing grounded, up-to-date travel information. Deployed **serverlessly** on Cloud Run and accessible via a **live dev UI**, it demonstrates how AI agents can reliably call external tools using open standards.

Meeting the Build Criteria

Track 2 - ADK + MCP Agent:
(1) Custom ADK Agent - built an **LlmAgent using Google ADK with McpToolset**, powered by **Gemini 2.5 Flash via Vertex AI**, with a system instruction for structured travel guidance.
(2) MCP Server — implemented a custom mcp.server.lowlevel.Server with 3 typed tools: get_country_info() calling REST Countries v3.1, get_wikipedia_summary() calling the Wikipedia REST API, and get_travel_advisory() **generating region-aware tips**.
(3) **Real-world API Integration** — agent fetches live data from two external APIs, grounding every response in authoritative facts.
(4) Deployment — fully **deployed on Google Cloud Run** (us-central1), containerised with Docker, built via Cloud Build, accessible at a live public URL. (5) Session Persistence — SQLite-backed aiosqlite session store through ADK's FastAPI server.

Opportunities

Differentiation: Unlike static travel apps or generic chatbots, Travel Guide Agent uses MCP as an open standard **("USB-C for AI")** to connect any AI agent to any tool — making the architecture modular and extensible. Most AI travel tools either hallucinate facts or require expensive proprietary data subscriptions; this agent fetches live, authoritative data at query time with no caching lag.

Problem Solved: Provides a single **conversational interface** that aggregates country facts, cultural context, and practical travel tips — all grounded in real-time API data, eliminating the need to visit 10+ sources manually.

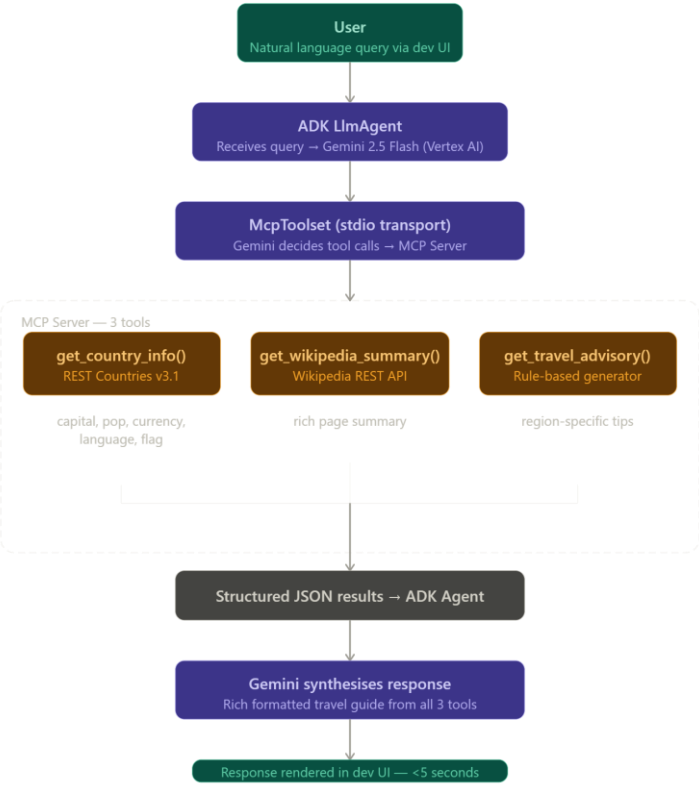
USP: **MCP-first architecture** means tools are p**lug-and-play** — new data sources (flights, weather, maps) can be added without touching agent code. Agent stays decoupled from APIs, enabling **rapid iteration and zero-hallucination** responses backed by live data.

List of features offered by the solution

1. Country Facts Lookup — Real-time data via REST Countries v3.1: capital, population, currency, language, region, and flag.
2. Wikipedia Summaries — Rich contextual summaries fetched from Wikipedia REST API for any country or topic.
3. Region-Aware Travel Advisories — Rule-based generator producing region-specific tips (visa reminders, currency advice, safety notes).
4. Natural Language Interface — Ask in plain English; Gemini 2.5 Flash orchestrates multi-tool calls and synthesises a single formatted response.
5. Session Persistence — SQLite-backed conversation memory via ADK's aiosqlite session store.
6. Serverless Deployment — Live on Cloud Run; zero cold-start issues, auto-scaling, publicly accessible dev UI.

7. MCP Tool Architecture — Tools are modular, typed JSON-schema MCP tools; easy to extend or swap without changing agent code. 8. Sub-5-second Response Time — All three tools called and response composed in under 5 seconds end-to-end.

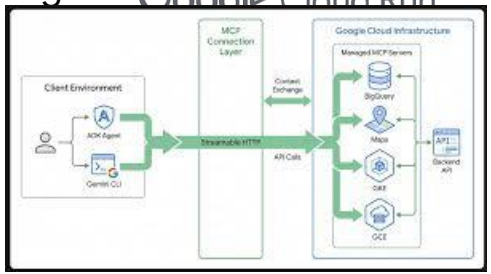
Process flow diagram or Use-case diagram



Technologies to be used in the solution

Agent Layer: Google ADK (Agent Development Kit) - agent orchestration framework.
Gemini 2.5 Flash — LLM powering the agent via Vertex AI.
Model Context Protocol (MCP) — open standard (by Anthropic, adopted by Google) for AI tool integration.
McpToolset — ADK component connecting agent to MCP server via stdio.
Backend : Python 3.12 — primary language.
FastAPI — web server via ADK's get_fast_api_app(). mcp.server.lowlevel.Server — MCP server implementation.
Wikipedia REST API — page summaries.
Region-based rule engine — travel advisory generation.
Infrastructure: Google Cloud Run (us-central1) — serverless container hosting.
Docker (python:3.12-slim) — containerisation.
Artifact Registry + Cloud Build — CI/CD pipeline.
Vertex AI — managed Gemini model endpoint.

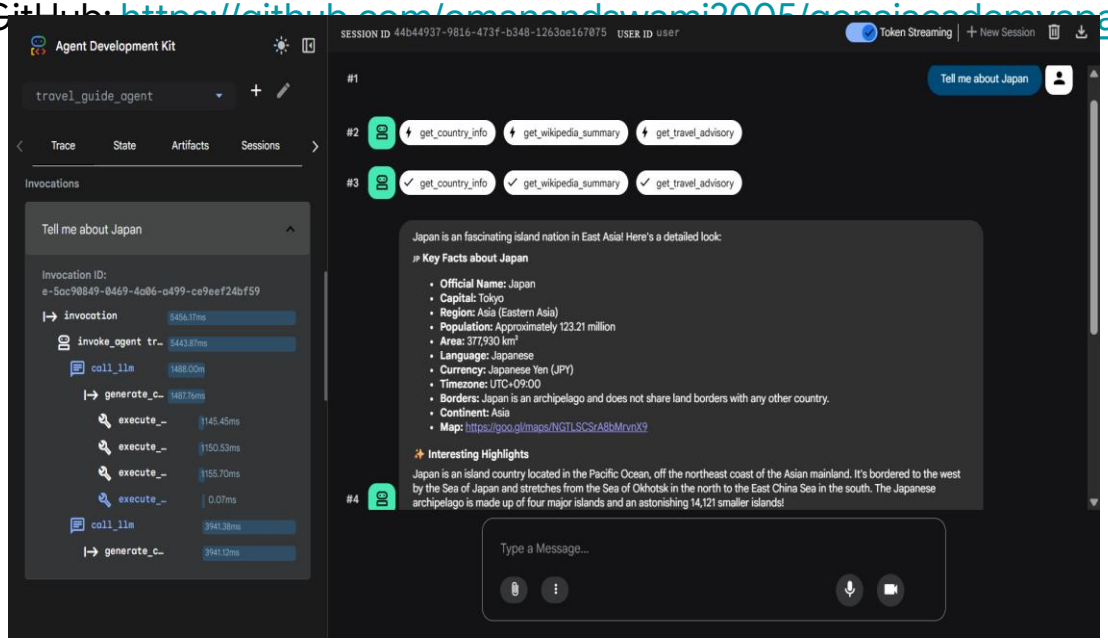
Google Cloud Run



Snapshots of the prototype

Live Prototype: https://travel-guide-agent-rnedu666qq-uc.a.run.app/dev-ui/?app=travel_guide_agent

GitHub: https://github.com/omanandswami2005/genaiacademyapac



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Thank you